MEETING LILY'S REQUIREMENTS

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Introduction

The aim of this project is to learn about how to meet customer requirements in software engineering. Here, I first interview Lily as a potential client. Next, I generate user stories, a user journey and a list of requirements. Finally, I create mockup wireframes for parts of the user journey using Figma.

The Requirements-Gathering Discussion

My conversation with Lily the Florist can be found using this link.

The User Stories

When creating the user stories below, I follow the INVEST-criteria:

- Independent: Each user story can be developed separately.
- Negotiable: Details can be refined during development.
- Valuable: Each user story addresses a specific pain point Lily mentioned.
- Estimable: Clear scope for each user story.
- Small: Each user story represents a manageable feature.
- Testable: Each user story has clear success criteria.

From the requirements-gathering discussion with Lily, I identify six types of prospective users. Below, I describe the pain points of these six hypothetical profiles from interaction with Lilys webpage: everyday gift buyer, event client, walk-in/immediate need, regular customer with standing orders, first-time customer and Lily herself. Finally, I make three user stories for mobile and desktop user needs.

Everyday Gift Buyer Persona

1. As an everyday gift buyer, I want to quickly browse ready-to-go bouquets with photos and prices, so that I can select a gift without needing consultation.
2. As a busy customer, I want to pay securely online with my credit card, so that I don't have to spend time on the phone giving card details.
3. As a gift buyer, I want to select available delivery time slots, so that I can ensure my flowers arrive when the recipient is available.
4. As a customer sending a gift, I want to add a personalized card message during checkout, so that my recipient knows who sent the flowers.

Event Client Persona

5. As an event client, I want to schedule initial consultations online, so that I can begin planning my wedding flowers at my convenience.
6. As an event planner, I want to view a portfolio of past designs and arrangements, so that I can communicate my vision clearly to Lily.
7. As an event client, I want to receive digital proposals and contracts, so that I can review and approve plans without needing in-person meetings.
8. As an event client, I want to track my project timeline and upcoming milestones, so that I know what's happening when leading up to my event.

Walk-in/Immediate Need Persona

9. As a walk-in customer, I want to see what fresh flowers and arrangements are currently available, so that I can make a quick purchase decision.
10. As a last-minute shopper, I want to reserve an arrangement for same-day pickup, so that I can secure flowers even when I can't come immediately.

Regular Customers with Standing Orders Persona

11. As a regular customer with standing orders, I want to manage my weekly arrangements online (skip weeks, change flowers, update delivery), so that I don't have to remember to call Lily each time.

First-time Customer Persona

12. As a first-time customer, I want to see what arrangements are currently available and in season with clear pricing, so that I can make an informed purchase decision without needing to call.
13. As a first-time customer, I want clear information about delivery areas, timing options, and flower care instructions, so that I have realistic expectations and can keep my flowers fresh.

Lily Persona

14. As Lily, I want real-time inventory tracking with automatic low-stock alerts, so that I don't oversell popular flowers during peak seasons.
15. As Lily, I want customers to select from available delivery time slots that automatically balance my schedule, so that I don't have drivers bunched up at the same times.

16. As Lily, I want to manage event client projects with all consultation notes, timelines, and contracts in one place, so that I can stay organized across multiple clients.
17. As Lily, I want route optimization for daily deliveries, so that my drivers can complete routes efficiently without backtracking.
18. As Lily, I want an automated system for managing standing orders with customer change notifications, so that I don't miss requests when I'm busy and can plan my weekly production accurately.
19. As Lily, I want the system to handle peak capacity of 300+ orders per day during holidays, so that the website doesn't crash during Valentine's Day or Mother's Day rushes.
20. As Lily, I want automated peak-season workflow management that prioritizes orders and balances delivery schedules, so that I can handle volume spikes without manual chaos.

As mobile and desktop users exhibit different interaction patterns and constraints, I also include mobile and desktop users needs to ensure system requirements address relevant technical considerations.

Mobile and Desktop User Needs

I infer the following user stories based on typical customer behavior:

21. As a busy customer on mobile, I want a simplified checkout process optimized for small screens, so that I can quickly order flowers while commuting or during work breaks.
22. As a customer browsing on desktop, I want detailed product galleries and comparison tools, so that I can thoroughly research arrangements for important events.
23. As a customer planning an event on desktop, I want to access consultation scheduling and project management tools, so that I can manage complex flower orders efficiently.

User Journey Map

I choose the event client journey, defined by user stories 5, 6, 7 and 8, for my user journey map.

From conversation with Lily, I synthesize the following workflow for the event client journey: Initial inquiry → Consultation → Design sketches → Proposal/contract → Timeline coordination → Event delivery

For this journey, I identify four phases and, in Figma, make the user journey shown in Figure 1 below:

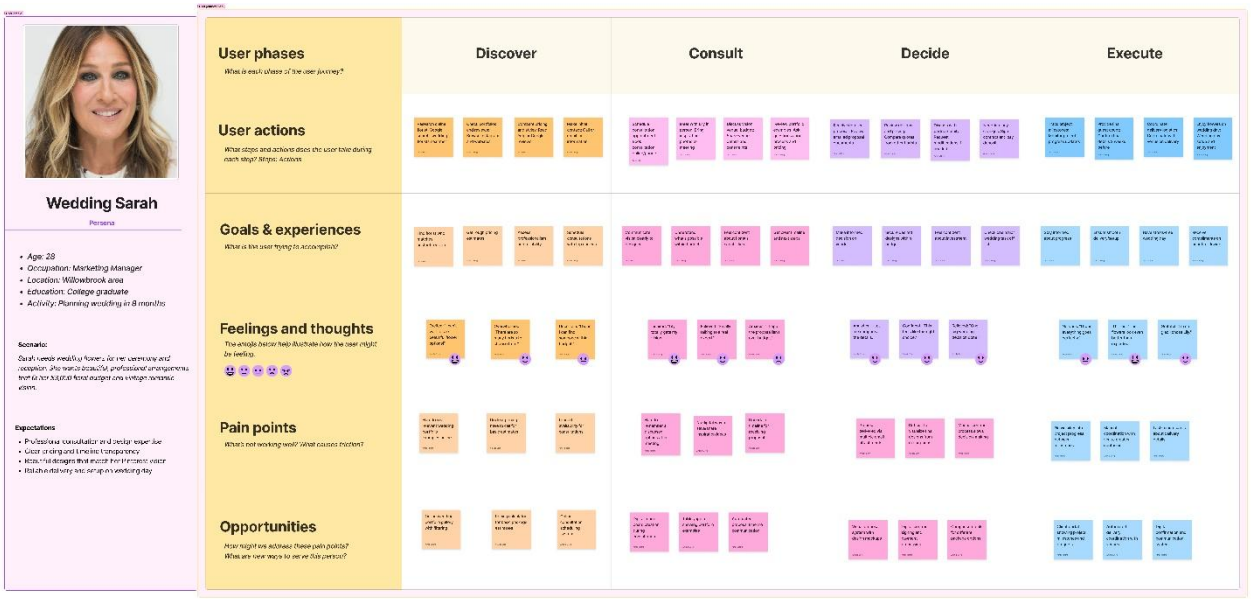


Figure 1: The user journey map for an event client journey exhibits details on a specific user persona, Sarah, and her four user phases, each with their own user actions, goals & experiences, feelings and thoughts, pain points and opportunities.

Requirements Specification

Functional Requirements

1. Customer Experience & Product Display

REQ-001 The system shall display current flower availability and seasonal status to customers with photos, prices, and last-updated timestamps. *(Stories 1, 9, 12)*

REQ-002 The system shall provide a comprehensive product gallery with high-resolution photos, detailed descriptions, and filtering options for arrangement types and price ranges. *(Stories 1, 9, 12, 22)*

REQ-003 The system shall include comparison tools allowing customers to view multiple arrangements side-by-side with pricing and feature differences. *(Story 22)*

REQ-004 The system shall provide an FAQ section addressing common questions about delivery areas, flower care, arrangement sizing, and pricing ranges. *(Stories 4, 12, 13)*

REQ-005 The system shall display clear pricing information for standard arrangements and provide size/price comparison guides for first-time customers. *(Stories 12, 13)*

2. Inventory Management

REQ-006 The system shall provide real-time inventory tracking that automatically updates quantities when flowers are received, used in arrangements, or sold individually. *(Story 14)*

REQ-007 The system shall generate automated low-stock alerts when flower quantities fall below predefined thresholds (configurable by flower type). *(Story 14)*

REQ-008 The system shall prevent customers from ordering flower types that are out of stock or below minimum arrangement requirements. *(Stories 1, 12, 14)*

3. Payment Processing

REQ-009 The system shall process credit card payments securely through encrypted online checkout, eliminating the need for phone-based card transactions. *(Stories 2, 12, 13)*

REQ-010 The system shall provide customers with immediate payment confirmation via email within 2 minutes of successful transaction completion. *(Stories 2, 13)*

REQ-011 The system shall store customer payment information securely using PCI DSS compliant tokenization for future purchases (with customer consent). *(Stories 2, 11)*

REQ-012 The system shall allow customers to save multiple payment methods and select their preferred option during checkout. *(Stories 2, 11)*

4. Delivery Management

REQ-013 The system shall present customers with available delivery time slots in 2-hour windows, preventing overbooking of delivery capacity. *(Stories 3, 13, 15)*

REQ-014 The system shall automatically balance delivery schedules to prevent more than 8 orders being assigned to the same 2-hour time slot. *(Story 15)*

REQ-015 The system shall generate optimized delivery routes grouped by geographic area to minimize driver backtracking and travel time. *(Story 17)*

REQ-016 The system shall clearly display delivery areas and estimated delivery times based on customer zip codes during the ordering process. *(Stories 3, 13)*

REQ-017 The system shall allow same-day pickup reservations for walk-in customers who cannot immediately visit the shop. *(Story 10)*

5. Event Management

REQ-018 The system shall provide online consultation scheduling with calendar integration showing Lily's available appointment slots. *(Stories 5, 23)*

REQ-019 The system shall maintain a digital portfolio gallery with wedding and event examples that customers can view and reference during consultations. *(Stories 6, 23)*

REQ-020 The system shall generate and deliver digital proposals with itemized pricing, design descriptions, and timeline milestones within 5 business days. *(Story 7)*

REQ-021 The system shall provide a client portal where event customers can track project progress, view timelines, and access contract documents. *(Stories 8, 23)*

REQ-022 The system shall store all consultation notes, design sketches, and client communications in a centralized project file accessible to Lily. *(Story 16)*

6. Standing Order Management

REQ-023 The system shall manage recurring weekly orders with automated production scheduling and customer billing cycles. *(Stories 11, 18)*

REQ-024 The system shall allow standing order customers to skip weeks, modify arrangements, or change delivery details through an online portal with 24-hour advance notice. *(Stories 11, 18)*

REQ-025 The system shall send weekly confirmation emails to standing order customers 48 hours before their scheduled arrangement preparation. *(Stories 11, 18)*

7. Personalization Features

REQ-026 The system shall allow customers to add personalized card messages during checkout with character limits and preview functionality. *(Story 4)*

Non-functional Requirements

8. Performance & Scalability

REQ-027 The system shall handle peak seasonal loads of up to 300 orders per day (3x normal capacity) without performance degradation or system failures. *(Stories 19, 20)*

REQ-028 The system shall implement automated peak-season workflow management that prioritizes urgent orders and automatically balances delivery schedules during volume spikes. *(Story 20)*

REQ-029 The system shall maintain response times under 3 seconds for all customer-facing pages during normal and peak usage periods. *(Stories 19, 21)*

REQ-030 The system shall process online payments within 30 seconds and provide immediate confirmation to customers. *(Stories 2, 21)*

9. Security & Data Protection

REQ-031 A user shall be allowed three login attempts before the system locks them out for 30 minutes to prevent unauthorized access attempts. *(Stories 2, 11 - account security)*

REQ-032 The system shall encrypt all customer data both in transit and at rest using industry-standard AES-256 encryption. *(Stories 2, 7, 11 - secure data handling)*

REQ-033 The system shall require customers to create accounts with passwords meeting minimum complexity requirements (8+ characters, mixed case, numbers). *(Stories 2, 11 - account security)*

REQ-034 Customer information shall be retained in the database for up to 7 years for business records, with automatic purging of inactive accounts older than 7 years (per business retention policy). *(Stories 11, 16 - data management)*

10. Mobile & Accessibility

REQ-035 The system shall provide a simplified, mobile-optimized checkout process with larger touch targets and minimal form fields for customers on smartphones. *(Story 21)*

REQ-036 The system shall be fully responsive and functional on mobile devices, tablets, and desktop computers across all major browsers. *(Stories 21, 22, 23)*

REQ-037 The system shall comply with WCAG 2.1 Level AA accessibility standards for users with disabilities. *(All stories - universal access)*

11. Integration & Backup

REQ-038 The system shall integrate with Lily's existing POS system for in-store sales reconciliation and unified customer records. *(Stories 9, 10 - walk-in integration)*

REQ-039 The system shall automatically backup all customer data, orders, and business records daily with 30-day retention and off-site storage. *(All stories - data protection)*

TRACEABILITY MATRIX

Requirement Category	Requirements	Related User Stories	Business Pain Point Addressed
Customer Experience	REQ-001 to REQ-005	Stories 1, 4, 9, 12, 13, 22	Customer confusion, lack of information
Inventory Management	REQ-006 to REQ-008	Stories 1, 12, 14	Manual tracking, stock-outs during peaks
Payment Processing	REQ-009 to REQ-012	Stories 2, 11, 12, 13	Phone payment delays and errors
Delivery Management	REQ-013 to REQ-017	Stories 3, 10, 13, 15, 17	Scheduling bunching, route inefficiency
Event Management	REQ-018 to REQ-022	Stories 5, 6, 7, 8, 16, 23	Manual project management, scattered info
Standing Orders	REQ-023 to REQ-025	Stories 11, 18	Informal order management
Personalization	REQ-026	Story 4	Gift customization needs
Performance	REQ-027 to REQ-030	Stories 19, 20, 21	Peak season system failures
Security	REQ-031 to REQ-034	Stories 2, 7, 11, 16	Data protection and account security
Mobile/Accessibility	REQ-035 to REQ-037	Stories 21, 22, 23	Cross-device functionality
Integration/Backup	REQ-038 to REQ-039	Stories 9, 10, All	System integration and data protection

Mockup Wireframes

I opt for making two mockup wireframes connected to the user journey. The goal here is to show the connection between the requirements and the wireframes. The first mockup wireframe is the landing page, and the second is the consultation booking process.

The Landing Page

The landing page is the entry point for multiple customer personas and implements inventory and user experience requirements (REQ-001, REQ-002, REQ-005, REQ-035, REQ-037). A simple mockup wireframe for this page is shown in Figure 2 below.

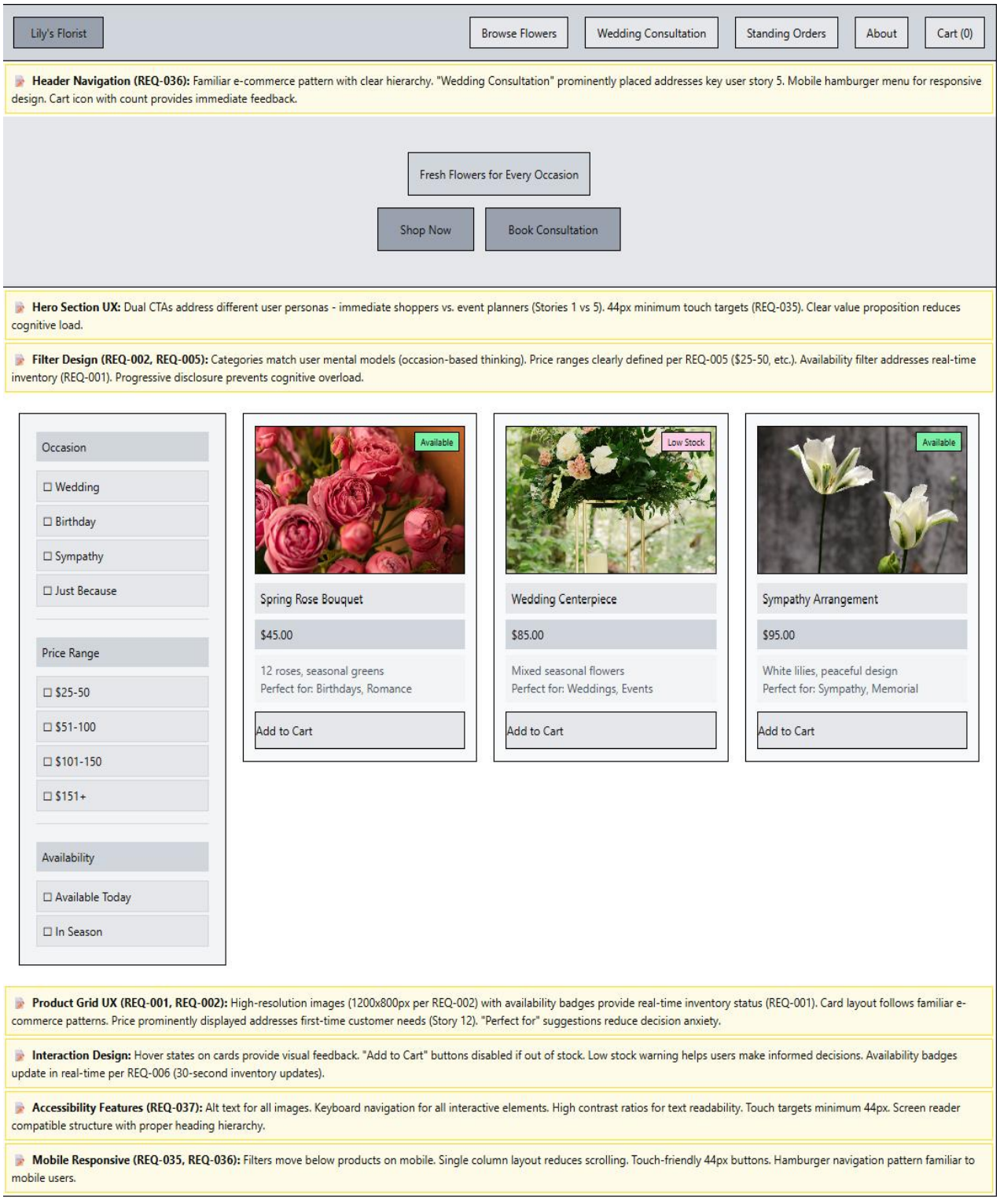


Figure 2: The landing page demonstrates product gallery layout, filtering functionality, and real-time inventory display. The design implements responsive navigation, accessibility features, and user-centered product discovery that address for example everyday gift buyer and first-time customer needs (Stories 1, 9, 12, 13). We see availability badges (REQ-001), high-resolution product images (REQ-002), clear pricing displays (REQ-005), and mobile-optimized touch targets (REQ-035).

The Consultation Booking Page

The consultation booking process represents an important touchpoint in the event client journey and implements multiple high-priority requirements (REQ-018, REQ-020, REQ-037) The wireframe for the this process is shown in Figure 3 below.

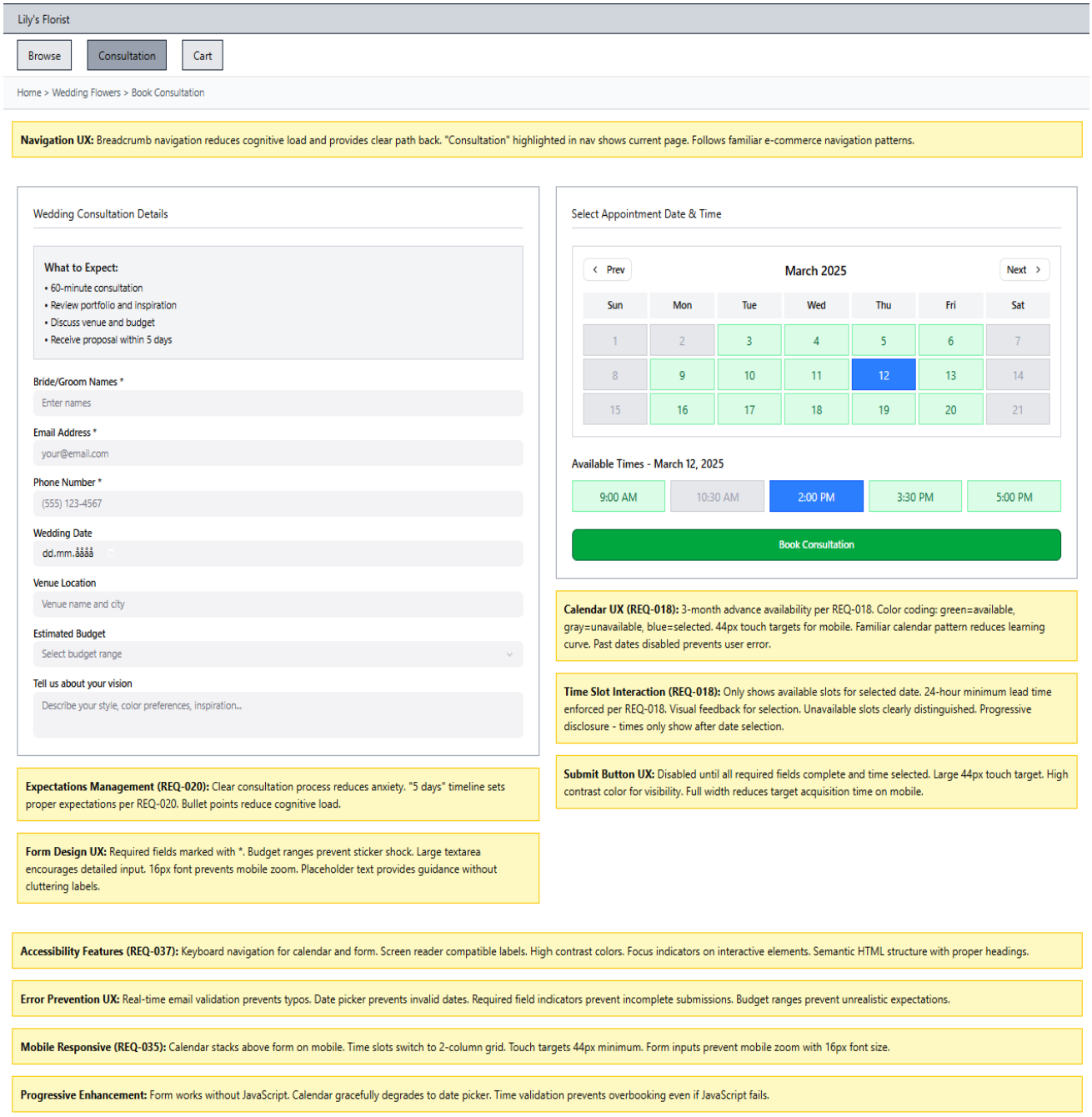


Figure 3: The consultation booking wireframe shows calendar selection, time slot booking, and consultation details form. The design demonstrates progressive disclosure principles, accessibility compliance (REQ-037), and mobile-responsive layout supporting the event client user journey (Stories 5, 23). Two important functionalities are the 3-month in advance booking availability (REQ-018) and 24-hour minimum lead time validation.

Final Remarks

Doing this project, I learnt how systematic user research and requirements management can address real operational challenges in software engineering projects. The interview with Lily revealed pain points around inventory management, delivery scheduling, and event coordination. These challenges directly shaped the journey map and the functional and non-functional system requirements. Finally, the wireframes showcase two webpage interfaces and their technical requirements. This work will improve business operations and customer experience for both Lily and her customers.